SATFORGE UAE 2026 · QUALIFIERS · PHASE 1

MARSAD-1

مرصد

A 3U CubeSat that watches the UAE's coastal waters for harmful algal blooms



Phase 1 Technical Report — Digital Design

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4waiz.github.io/marsad-1-dashboard
Live mission dashboard

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1 Executive summary

MARSAD-1 (Arabic *marsad*, "observatory") is a 3U CubeSat designed by Team Kanban to detect and map harmful algal blooms ("red tides") and chlorophyll-a concentration along the UAE coastline of the Arabian Gulf and the Gulf of Oman. It flies in a 550 km sun-synchronous orbit, images UAE coastal waters with a 5 MP camera and an 11-channel spectral sensor, computes a chlorophyll index on board, and sends compact bloom alerts to a ground station in Abu Dhabi over a 433 MHz LoRa link.

All Phase 1 design budgets close with margin on the SpacePoint Madar mission designer, using a bill of materials of 17 real, purchasable components:

+2.68 W
Power margin · 6.60 W generated vs 3.92 W peak load

0.845 kg
Mass · of 4.0 kg for a 3U (79% margin)

1,420 AED
Cost · of 3,000 AED budget

75 MB/day
Data captured · 2 GB onboard storage

Phase 1 deliverables produced by the team:

- Individual and team Madar mission designs (setup, components, CONOPS, data, power, mass and cost budgets) — all phases cleared.
- A Tinkercad Arduino circuit that simulates the satellite's power, thermal, payload and communications mode logic and prints the same JSON telemetry the dashboard reads.
- A Tinkercad 3D model of the 3U spacecraft built to the CubeSat Design Specification envelope.
- A live interactive mission dashboard published on GitHub Pages, and a competition poster.

2 Problem & motivation

Harmful algal blooms occur when certain species of phytoplankton multiply rapidly. Some species discolour the water red or brown ("red tides"), deplete dissolved oxygen as the bloom decays, or produce toxins. The warm, shallow and semi-enclosed waters of the Arabian Gulf and the Gulf of Oman are susceptible to such events, and large blooms have been recorded in the region.

For the UAE the impact is direct:

- **Desalination** — much of the country's drinking water comes from seawater desalination. Dense blooms can clog intake screens and foul pre-treatment filters and reverse-osmosis membranes, and plants have had to reduce output or shut down during severe events.
- **Fisheries and aquaculture** — oxygen depletion and toxins can cause fish kills and force closures.
- **Coastal tourism and public health** — discoloured water and dead fish on beaches, and potential toxin exposure.

> **The gap.** Blooms can develop and drift over days. Operators benefit most from *early, frequent, local* information: where a bloom is, how dense it is and whether it is moving toward an intake. Large Earth-observation missions provide excellent ocean-colour data but are not tailored to a single coastline. A dedicated, low-cost CubeSat that revisits UAE waters daily and pushes simple alerts to operators complements them.

3 Mission objectives & success criteria

Item	Statement
Primary objective	Detect and map harmful algal blooms (red tides) and chlorophyll-a concentration along the UAE coastline of the Arabian Gulf and Gulf of Oman using multispectral imaging, to give early warning to desalination plants, fisheries and coastal authorities.
Success criterion	Downlink at least one usable, cloud-free multispectral image of UAE coastal waters per day and deliver a chlorophyll-a bloom map to the ground team within 24 hours, for 6 consecutive months.
Minimum lifetime	12 months.
Mission tasks	Nadir-pointed multispectral imaging during sunlit passes over the Gulf, onboard compression and bloom detection (NDCI), and downlink of alerts and data over UAE ground-station passes, with UHF telemetry and commanding.
Secondary objective	Demonstrate a low-cost (< 3,000 AED bill of materials) student CubeSat architecture with onboard data reduction.

4 Orbit & concept of operations

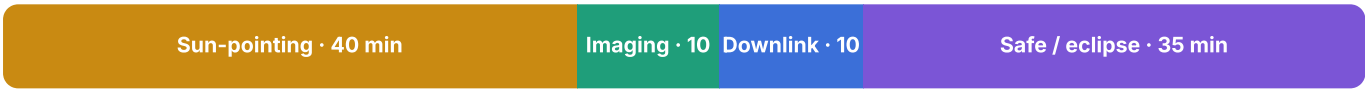
Orbit

Type	Sun-synchronous (SSO)
Altitude	550 km
Inclination	97.6°
Local time of descending node	10:30
Period	95 min
Orbits per day	15
Ground station	Abu Dhabi (24.45° N, 54.38° E), 10° elevation mask

Why SSO at 550 km

A sun-synchronous orbit crosses the UAE at nearly the same local solar time every day, giving consistent illumination for ocean-colour measurements — important when comparing bloom maps day to day. A mid-morning node (10:30) avoids the strongest afternoon convection and sun glint. 550 km is a common rideshare altitude with a multi-year natural decay time, well beyond the 12-month mission.

Concept of operations — one 95-minute orbit



Timeline — nominal mode allocation per orbit (as entered in the Madar CONOPS phase; total 95 min = orbit period).

Mode	Min / orbit	Active components	Load (W)
Sun-pointing (charging)	40	OBC, IMU, magnetorquers, solar array, EPS (battery, MPPT/BMS, bucks, INA219), temp sensors, structure switches	1.64
Nadir / payload imaging	10	Sun-pointing set + OV5640 camera, AS7341 spectral sensor, GPS, microSD	2.52
Ground-station pass	10	Sun-pointing set + GPS, SX1278 LoRa radio, antenna, microSD	2.45
Safe / eclipse	35	OBC, IMU, EPS, temp sensors, **battery heater**; magnetorquers and payload off	1.88

Mode logic. The flight software enters *safe* whenever the satellite is in eclipse or battery state of charge drops below 20%. In sunlight it holds a sun-pointing, charging attitude; when the ground track crosses the UAE target box and charge is above 60% it switches to nadir imaging; inside the Abu Dhabi visibility circle it enters the ground-pass mode and downlinks queued alerts. The battery heater operates with hysteresis (on below 5 °C, off above 10 °C).

5 System architecture & subsystems

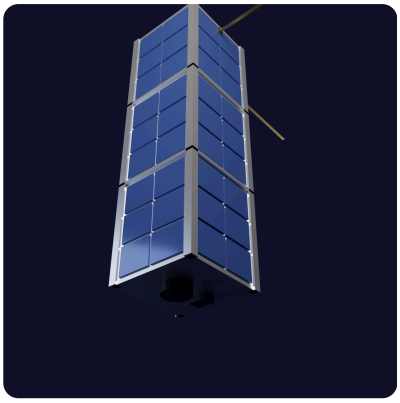


Figure 1 — Model seen from below: cells, nadir camera and spectral window.

CDHS — command & data handling

An **ESP32-S3** (dual-core 240 MHz, 8 MB PSRAM, 16 MB flash) runs the flight-software state machine, schedules imaging, reads all sensors over I²C/SPI, and formats telemetry. A **32 GB microSD** stores raw images, spectra and logs between ground passes. The ESP32-S3's camera interface (DVP) connects the imager directly.

Payload — bloom detection

The **OV5640** 5 MP camera captures RGB context imagery for visual confirmation. The **AS7341** 11-channel spectral sensor (350–1000 nm) measures water-leaving light in narrow bands. Chlorophyll-rich bloom water reflects more strongly in the red-edge (~708 nm) relative to the chlorophyll absorption band (~665 nm), which is captured by the Normalised Difference Chlorophyll Index:

$$NDCI = (R_{708} - R_{665}) / (R_{708} + R_{665})$$

NDCI is computed on board. Pixels or samples above a threshold are tagged, and a compact alert packet (location, index, time) is queued for the next ground pass. Full images are kept and downlinked selectively. On the AS7341 the closest available bands are used (F8 ≈ 680 nm and the NIR channel as the red-edge proxy); calibration against reference data is planned for the next phase.

ADCS — attitude determination & control

An **ICM-20948** 9-DoF IMU (gyro, accelerometer, magnetometer) provides attitude determination; a **3-axis magnetorquer set** driven by a DRV8833 H-bridge detumbles the spacecraft (B-dot) and holds coarse sun/nadir pointing. A **u-blox NEO-M8N** GPS gives position for geotagging images and precise time for imaging schedules.

COMMS — communications

A **Semtech SX1278 LoRa** transceiver at 433 MHz (up to +20 dBm) carries telemetry, commands and bloom alerts. LoRa's long-range, low-rate modulation suits small alert packets. A **deployable spring-steel tape monopole** is released by a burn-wire after separation.

EPS — electrical power

Six 1.1 W body-mounted solar cells (6.6 W) charge a **2S Li-ion 18650 pack** (7.4 V, 3.4 Ah, ≈ 25 Wh) through a **CN3791 MPPT charger with 2S BMS**. **MP1584 buck converters** provide regulated 5 V and 3.3 V rails, with load switching from the OBC, and an **INA219** monitors bus voltage and current.

Thermal

A **BME280** and **DS18B20** probes monitor internal, battery and panel temperatures. A **1 W Kapton heater pad** on the battery keeps the cells above 0 °C during the 35-minute eclipse.

Structure

A **3U aluminium 6061 frame** (100 × 100 × 340.5 mm envelope, four 8.5 mm rails, 1U separator rings) with a PC/104-style board stack, following the CubeSat Design Specification, plus two deployment (kill) switches.

6 Budgets

All budgets were computed and locked in the SpacePoint Madar mission designer. Power per component follows the designer's convention `P = V × I` at the operating point.

6.1 Power

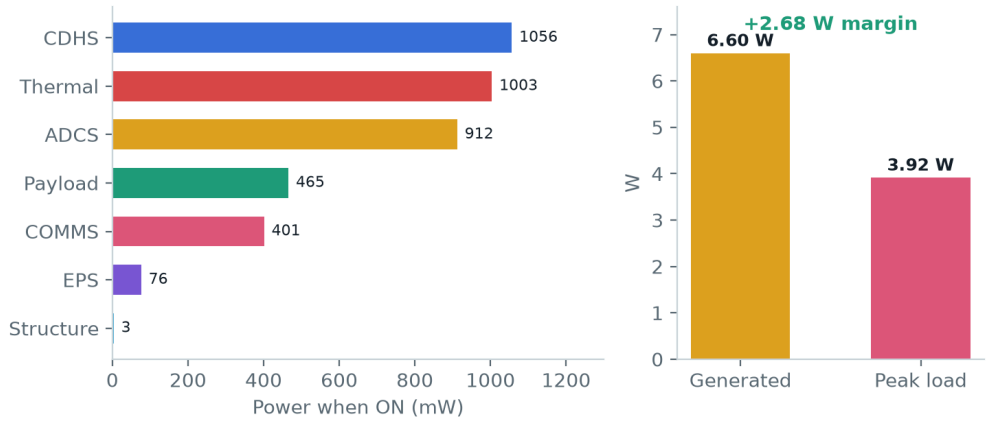


Figure 2 — Left: power drawn by each subsystem when ON. Right: generated power vs total load.

Mode	Load (W)	Min	mWh / orbit
Sun-pointing	1.64	40	1,094
Nadir imaging	2.52	10	420
Ground pass	2.45	10	409
Safe / eclipse	1.88	35	1,098
Total		**95**	**3,020**

Sizing. Total load with every component on is **3.917 W** (3,916.5 mW). With 1.1 W cells, ⌈3.917 / 1.1⌉ = 4 cells would just cover it; we selected **6 cells = 6.60 W**, a **+2.68 W (68%)** margin that covers cosine losses of body-mounted cells and degradation.

Energy. 3.02 Wh is used per orbit (45.3 Wh per day). The 25 Wh battery covers the 35-minute eclipse load (≈ 1.1 Wh) with a shallow depth of discharge (≈ 4–5%), which favours cycle life.

6.2 Mass & volume

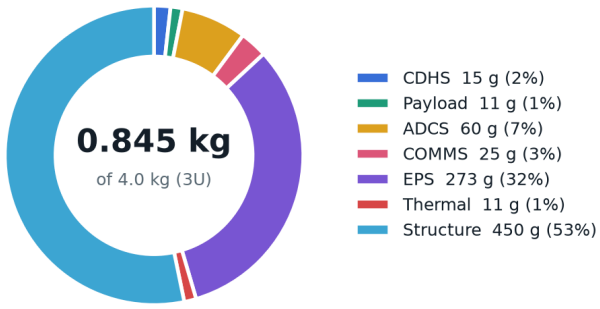


Figure 3 — Mass by subsystem (total 845 g).

Mass: 0.845 kg against the 4.0 kg 3U limit — a **3.16 kg (79%) margin**. The structure (450 g) and EPS (273 g, mostly the six panels and battery) dominate.

Volume: 500.6 cm³ of component volume against 3,000 cm³ available (the frame is counted by its material footprint, not its envelope). The large margin leaves room for harnessing, a larger battery, or a second payload in later phases.

6.3 Data

Source	Type	Size / meas.	Meas. / min	Active min	Per day	Handling
OV5640 camera	Image	500 KB	1	10	75.0 MB	Stored; selected scenes downlinked
AS7341 spectral	Science	0.05 KB	60	10	450 KB	Stored + sent
ESP32-S3 OBC	Telemetry	0.2 KB	1	95	285 KB	Stored + sent
ICM-20948 IMU	Telemetry	0.03 KB	6	95	257 KB	Stored
NEO-M8N GPS	GPS/Nav	0.1 KB	6	20	180 KB	Stored + sent
SX1278 LoRa	TT&C	0.25 KB	6	10	225 KB	Sent
EPS / thermal / other	Housekeeping	0.01–0.05 KB	1–2	10–95	small	Stored (key values sent)
Total generated: 5.00 MB per orbit					**75.0 MB**	**74.78 MB stored · 1.42 MB sent**

> **Design driver.** The satellite captures about 50× more data per day than the LoRa link can send. A 2 GB card holds roughly 27 days of raw imagery. MARSAD-1 therefore processes images on board and sends *results* (kilobyte bloom alerts and thumbnails) instead of raw images, and keeps only bloom-flagged scenes.

6.4 Cost & bill of materials

#	Component	Subsystem	Qty	V	mA	P (mW)	Mass (g)	Cost (AED)	Vendor
1	ESP32-S3 on-board computer	CDHS	1	3.3	240	792	10	60	DigiKey
2	OV5640 5 MP camera	Payload	1	3.3	140	462	8	45	Arducam
3	AS7341 11-channel spectral sensor	Payload	1	3.3	1	3.3	3	65	Adafruit
4	ICM-20948 9-DoF IMU	ADCS	1	3.3	4	13.2	3	55	SparkFun
5	3-axis magnetorquers + DRV8833	ADCS	1	5	150	750	45	120	DigiKey
6	u-blox NEO-M8N GPS	ADCS	1	3.3	45	148.5	12	75	u-blox
7	SX1278 LoRa 433 MHz transceiver	COMMS	1	3.3	120	396	5	35	DigiKey
8	Deployable 433 MHz tape antenna	COMMS	1	5	1	5	20	40	Amazon.ae
9	Solar cell, 6 V 1 W	EPS	6	6	1	6	150	150	Amazon.ae
10	2S Li-ion 18650 pack 7.4 V 3.4 Ah	EPS	1	7.4	1	7.4	95	90	DigiKey
11	CN3791 MPPT charger + 2S BMS	EPS	1	7.4	5	37	10	40	AliExpress
12	MP1584 buck converters (5 V, 3.3 V)	EPS	2	7.4	3	22.2	16	40	DigiKey
13	INA219 current/voltage monitor	EPS	1	3.3	1	3.3	2	25	Adafruit
14	MicroSD storage module (32 GB)	CDHS	1	3.3	80	264	5	35	Amazon.ae
15	BME280 + DS18B20 temperature sensors	Thermal	1	3.3	1	3.3	6	30	SparkFun
16	Kapton battery heater pad (1 W)	Thermal	1	5	200	1000	5	35	Amazon.ae
17	3U Al-6061 frame + PC/104 stack	Structure	1	3.3	1	3.3	450	480	CubeSatShop
	Total (17 line items)					**3,916.5**	**845**	**1,420**	

Cost: 1,420 AED against the 3,000 AED budget (1,580 AED margin). The frame is the most expensive item (480 AED); the margin can absorb a flight-grade antenna, spare parts and test equipment.

7 Prototype: Tinkercad circuit

Before buying hardware we simulated the flight computer's sensing and mode logic in Tinkercad Circuits with an Arduino Uno standing in for the ESP32-S3. Circuit: tinkercad.com/things/cY2kWDnTdZB (*Team Kanban – Circuit Design - Awaiz Ahmed*).

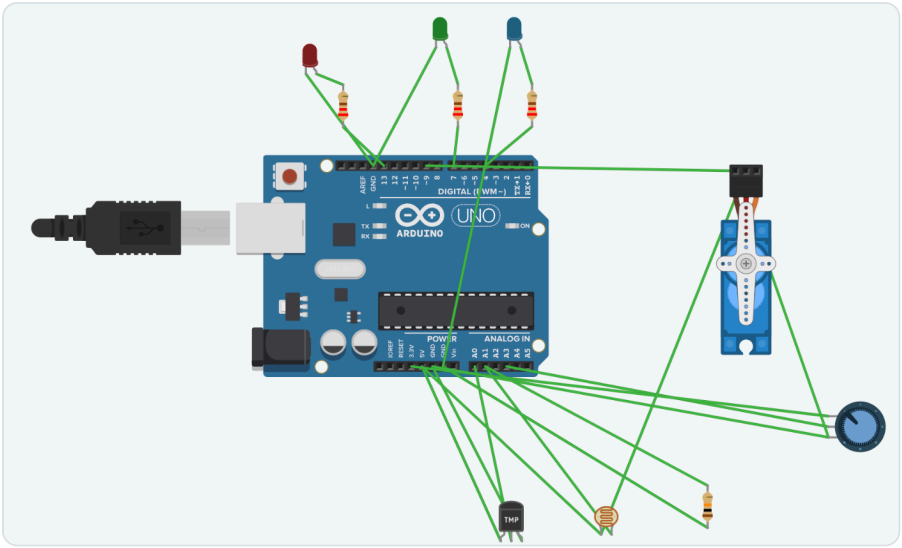


Figure 4 — Tinkercad circuit: TMP36, photoresistor divider, potentiometer, three status LEDs and the antenna-deployment servo.

Pin	Part	Represents
`A0`	TMP36	Battery temperature
`A1`	Photoresistor + 10 kΩ pull-down	Sun sensor (sunlit / eclipse)
`A3`	Potentiometer (3.3 V)	Battery voltage → state of charge
`D13`	Red LED + 220 Ω	Battery heater
`D7`	Green LED + 220 Ω	Payload imaging (nadir)
`D4`	Blue LED + 220 Ω	UHF downlink / ground pass
`D9`	Micro servo	Antenna deployment

Behaviour verified in simulation:

- Boot message, then the antenna servo rotates 0° → 90° five seconds after power-up (separation timer).
- Mode selection: *safe* if SoC < 20% or no sunlight; *ground pass* for 2 s of every 10 s once the antenna is out; *nadir imaging* if SoC > 60%; otherwise *sun-pointing*.
- Heater LED with 5 °C / 10 °C hysteresis on the TMP36 reading.
- Imaging and downlink LEDs follow the active mode.

Telemetry format

The sketch prints one JSON line per second over serial (9600 baud in Tinkercad, 115200 on the ESP32-S3). The dashboard's Web Serial connection reads the same keys:

```
MARSAD-1 boot - Team Kanban
{"soc":100.0,"v":8.40,"tbat":24.8,"sun":54,"mode":"nadir","heater":false,"antenna":false}
EVENT antenna deployed
{"soc":100.0,"v":8.40,"tbat":24.8,"sun":54,"mode":"gs","heater":false,"antenna":true}
```

Source: `firmware/tinkercad_prototype/tinkercad_prototype.ino`; ESP32-S3 flight sketch with INA219, BME280, DS18B20, AS7341 and NDCI: `firmware/marsad_telemetry/marsad_telemetry.ino`.

8 3D design

The spacecraft was modelled to the CubeSat Design Specification 3U envelope and assembled in Tinkercad 3D: tinkercad.com/things/4Qi0WBiqTus (*Team Kanban – 3D Design - Awaiz Ahmed*). Parts were generated parametrically (`cad/make_cubesat.py`), imported as colour-separated STL parts and rendered in Blender.

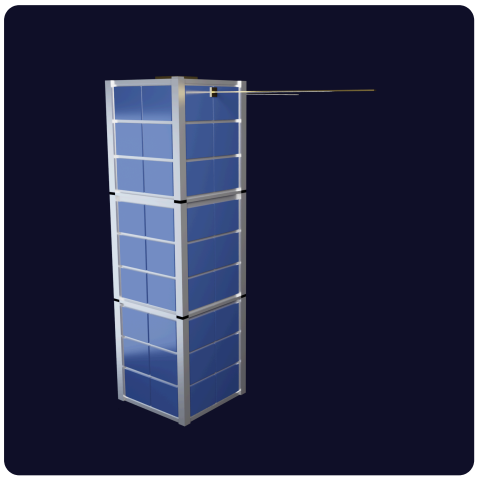


Figure 5 — Blender render of the 3D model.

Feature	Detail
Envelope	100 × 100 × 340.5 mm (3U)
Frame	Four 8.5 mm corner rails, top and bottom plates, two 1U separator rings
Solar cells	72 body-mounted cell areas (2 × 3 per unit, 4 faces, 3 units)
Nadir face	Camera lens barrel and spectral-sensor window
Zenith face	GPS patch antenna, two deployment kill switches
Antennas	Two deployed 433 MHz tape elements (170 mm) with hinge blocks
Colours	Silver (Al-6061), deep blue (cells), gold (antennas), black (optics)

9 Ground segment & dashboard

The mission dashboard is live at 4waiz.github.io/marsad-1-dashboard and is open source.

> **Honesty note.** Until the physical prototype is connected, telemetry on the dashboard is **simulated** from the design budgets and orbit model, and the bloom map uses a **seeded demonstration field**. Both are labelled as such on the page.

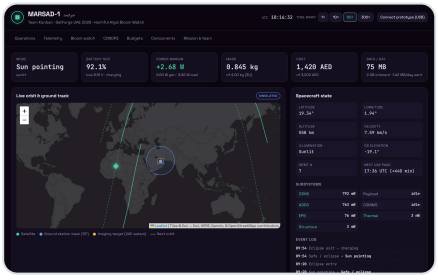


Figure 6 — Operations view: live ground track, spacecraft state, subsystem status and event log.

- **Operations** — ground track from a sun-synchronous orbit propagator, eclipse and Abu Dhabi pass prediction, current CONOPS mode, subsystem status and an event log. A time warp (1×–300×) lets a full orbit be watched in minutes.
- **Telemetry** — battery state of charge, generated vs consumed power, internal/battery/panel temperatures and onboard storage.
- **Bloom watch** — a chlorophyll-a (NDCI) map of UAE waters with watch/warning status for desalination plants and fisheries.
- **Prototype link** — a *Connect prototype (USB)* button uses the browser's Web Serial API (Chrome/Edge) to read the JSON telemetry from the hardware and switch the panels from simulated to live data.

10 Risks & mitigations

Risk	Impact	Mitigation
Cloud cover hides the target	Medium	15 orbits/day and daily revisits; multi-day composites; bloom alerts only from cloud-free samples; success criterion requires one usable image per day, not per pass.
Downlink capacity (1.42 MB/day)	High	Onboard NDCI processing; send alerts and thumbnails, not raw imagery; prioritise bloom-flagged scenes; option for additional ground stations.
Radiation effects on COTS electronics (latch-up, bit flips)	High	Watchdog and safe mode, INA219 over-current detection with load power-cycling, CRC on stored data, periodic reboots.
Battery too cold in eclipse	Medium	Kapton heater with hysteresis; battery temperature monitoring; heater energy already in the power budget.
Coarse pointing with magnetorquers only	Medium	Wide field-of-view imaging; GPS + IMU geotagging; image only when attitude is stable.
Antenna fails to deploy	Medium	Redundant burn-wire activation attempts; beacon via stowed antenna at reduced range.
NDCI accuracy without calibration	Medium	Ground calibration against reference ocean-colour data; conservative thresholds; human-in-the-loop confirmation with RGB images.
Schedule for the physical prototype	Medium	Parts chosen from stocked distributors; simulation-first development; tasks split by subsystem owner.

11 Next steps toward the final qualifier

- **Physical prototype** on an ESP32-S3 with the INA219, BME280, DS18B20 and AS7341 breakouts, running `marsad_telemetry.ino` and streaming to the dashboard over USB.
- **Multisim simulation** of the EPS (solar input, MPPT charging, buck regulation) and the heater driver, with a recorded walkthrough.
- **Testing** — sensor calibration, mode-switch tests with a lamp as the "sun", thermal test of the heater loop, LoRa range test between two modules, battery charge/discharge cycles.
- **Link budget** for the 433 MHz downlink at 550 km and the Abu Dhabi ground station.
- **Algorithm work** — NDCI thresholds and calibration using public ocean-colour reference data.
- **Final deliverables** — prototype video, technical report, final dashboard and recorded presentation.

12 Key learnings

- **Budgets drive the design.** The power budget set the array at six cells and made the eclipse heater a visible line item.
- **Downlink, not mass, is the bottleneck.** Capturing 75 MB/day while sending 1.42 MB/day forced onboard processing.
- **Mass is easy in a 3U; layout, cabling and heat paths are not.** We used under a quarter of the mass limit but must still plan volume and thermal paths.
- **Simulate first.** Tinkercad exposed wiring mistakes and let us test the mode logic before buying parts.
- **One source of truth.** A shared live dashboard kept four subsystem owners working from the same numbers.

A Appendix: links & repository

Item	Link
Live dashboard	https://4waiz.github.io/marsad-1-dashboard/
Source repository	https://github.com/4waiz/marsad-1-dashboard
Tinkercad circuit	https://www.tinkercad.com/things/cY2kWDnTdZB
Tinkercad 3D design	https://www.tinkercad.com/things/4Qi0WBiqTus
Prototype firmware	`firmware/tinkercad_prototype/` · `firmware/marsad_telemetry/`
CAD & render scripts	`cad/make_cubesat.py` · `cad/render_cubesat.py`

The cover image is an AI-generated artist's impression. All other figures are produced from the team's own design data, models, circuit and dashboard.